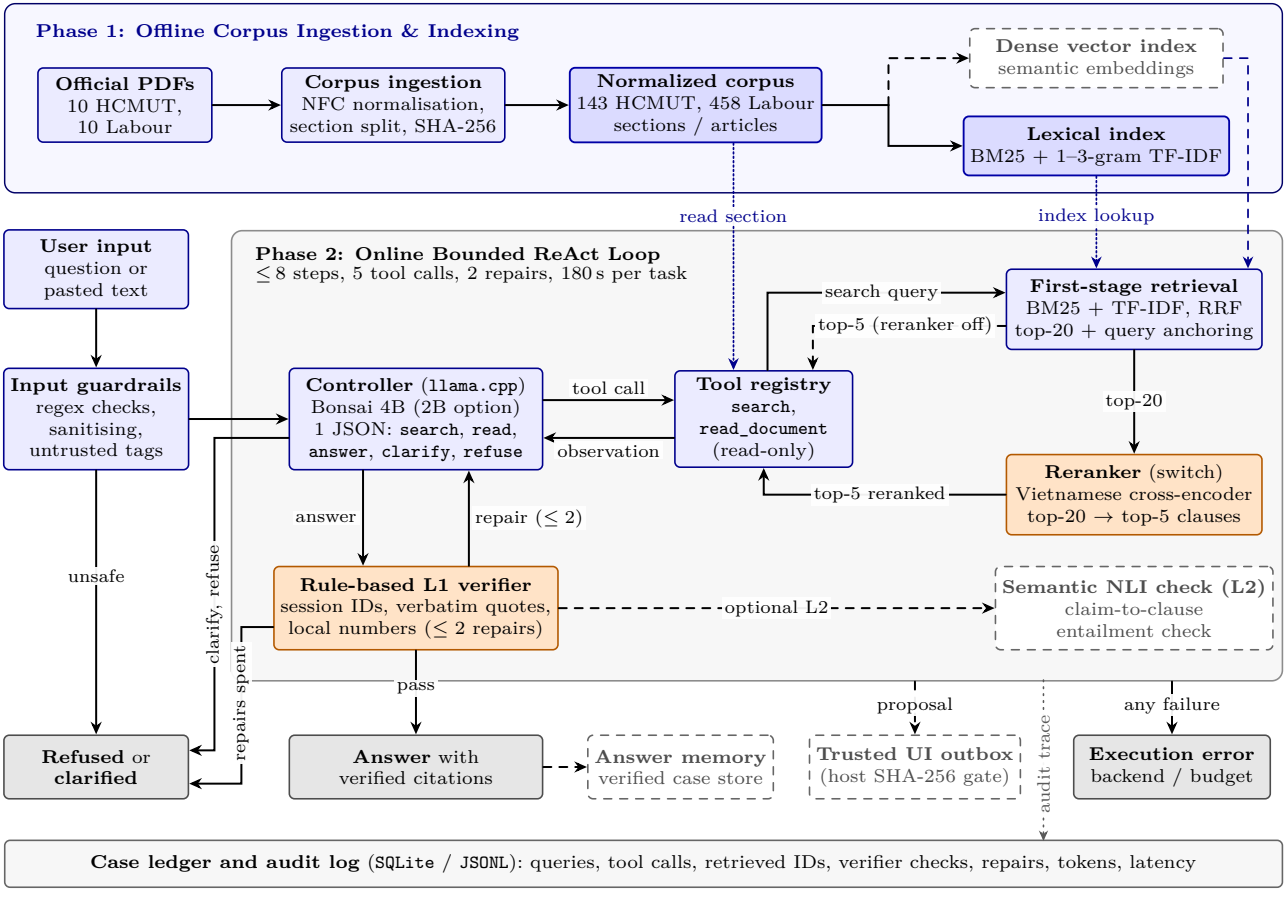

Phase 1: Offline Corpus Ingestion & Indexing
Official PDFs
10 HCMUT, 10 Labour
Corpus ingestion
NFC normalisation, section split, SHA-256
Normalized corpus
143 HCMUT, 458 Labour sections / articles
Dense vector index
semantic embeddings
Lexical index
BM25 + 1–3-gram TF-IDF
read section
index lookup
User input
question or pasted text
Phase 2: Online Bounded ReAct Loop
≤ 8 steps, 5 tool calls, 2 repairs, 180 s per task
search query
top-5 (reranker off)
First-stage retrieval
BM25 + TF-IDF, RRF
top-20 + query anchoring
top-20
Input guardrails
regex checks, sanitising, untrusted tags
Controller (llama.cpp)
Bonsai 4B (2B option)
1 JSON: search, read, answer, clarify, refuse
tool call
observation
Tool registry
search, read_document
(read-only)
Reranker (switch)
Vietnamese cross-encoder
top-20 → top-5 clauses
top-5 reranked
answer
repair (≤ 2)
unsafe
clarify, refuse
repairs spent
Rule-based L1 verifier
session IDs, verbatim quotes, local numbers (≤ 2 repairs)
optional L2
Semantic NLI check (L2)
claim-to-clause entailment check
pass
proposal
audit trace
any failure
Refused or clarified
Answer with verified citations
Answer memory
verified case store
Trusted UI outbox
(host SHA-256 gate)
Execution error
backend / budget
Case ledger and audit log (SQLite / JSONL): queries, tool calls, retrieved IDs, verifier checks, repairs, tokens, latency